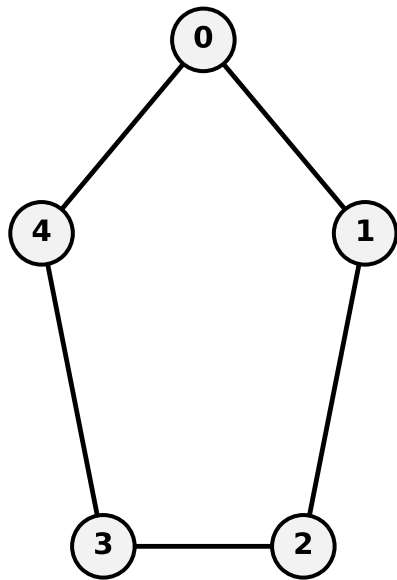
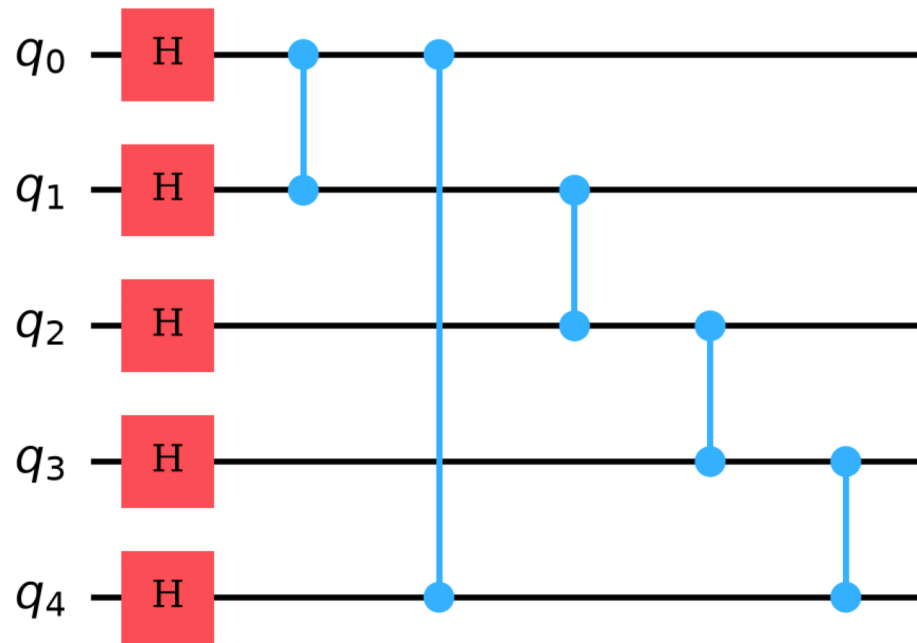


(a) Five-cycle graph C_5



(b) Graph-state preparation for C_5



(c) Rank-4 subgroup of S_{C_5}

$$S_{C_5} = \langle K_0, \dots, K_4 \rangle$$

(rank 5)



select rank-4 subgroup

$$S_0 = XZZXI$$

$$S_1 = IXZZX$$

$$S_2 = XIXZZ$$

$$S_3 = ZXIXZ$$



$$[[5, 1, 3]]$$

$$X_L = IIXYX$$

$$Z_L = IIZXZ$$